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Professional Summary

M.Sc. Computer Science candidate at JMU Würzburg with dual expertise in **Aerospace Informatics (Machine Learning)** and **HCI/VR/XR development**. Pursuing a **Master's thesis on Neural-Network-based Initial Orbit Determination**, building on 2 aerospace Praktikums (angles-only IOD via MLP/LSTM; CR3BP orbit prediction) and 3 VR/HCI Praktikums (Meta Quest 3 game dev; multimodal interaction).

Projects and Experience

Praktikum – VR Game Dev: Forest Siege 🎮 📺

Nov 2025 – Mar 2026

Research Assistant · JMU Würzburg

- Solo Unity (C#) VR game for Meta Quest 3 — stable 72fps, physics-based mechanics; structured playtest with 8 participants.
- Built **session data logging pipeline** (player movement & events) as ML research dataset for adaptive behaviour modelling.

3DUI – Skyline Sprint: VR Parkour Game 🎮 📺

Oct 2023 – Feb 2024

3DUI Course · JMU Würzburg

- Developed immersive VR parkour game in Unity (C#) with 3+ custom locomotion mechanics (dynamic jump, wall-run, rooftop traversal) using XR Interaction Toolkit — strong academic evaluation.

Praktikum – Aerospace: Angles-Only IOD via Neural Networks 🧠

May 2025 – Sep 2025

Research Assistant · JMU Würzburg | Foundation of Master's Thesis

- Built & benchmarked **MLP and LSTM (PyTorch)** for IOD of ESA MASTER space objects from angles-only data (Az, El, Time); MLP MAE 45–235 km vs. Gauss/Laplace 55–124 km.
- Full pipeline: Optuna tuning, Z-score/min-max normalisation, outlier detection, MAE/RMSE on 6D state vectors.

Praktikum – Aerospace: Pose Estimation & Orbit Prediction (CR3BP) 🧠

Oct 2025 – Feb 2026

Research Assistant · JMU Würzburg

- Trained **MLP (PyTorch)** to predict periodic orbit ICs from Jacobi constants in Earth-Moon CR3BP; feature engineering, Optuna tuning & regularisation across orbit families (Lyapunov, Halo, Axial).

MMI – Multimodal VR Interaction 🎮 📺

Nov 2023 – Feb 2024

MMI Course · JMU Würzburg

- Unity VR system with real-time speech + probabilistic **FSM multimodal fusion** (voice + pointing + gesture object control).

Praktikum – Computer Vision: Image Enhancement & Model Robustness

Jan – May 2025

Research Assistant · JMU Würzburg

- Implemented **InstructIR** pipelines; benchmarked CV robustness via PSNR/SSIM across noise, blur & compression.

Web Developer – AED-IT Software

Jun – Nov 2022

Rajkot, India · Front-end & Back-end

Delivered responsive cross-browser sites (HTML/CSS/JS); collaborated through full project lifecycle.

Education

M.Sc. Computer Science — Thesis: NN-based Initial Orbit Determination

Oct 2023 – Present · Würzburg, DE

Julius-Maximilians-Universität Würzburg

B.Eng. Computer Science and Engineering

2018 – 2022 · Vadodara, IN

Gujarat Technological University

Technical Skills

- **Game/XR:** Unity (C#), XR Interaction Toolkit, Meta Quest 3 SDK, VR/AR Development, Physics Systems
- **AI/ML & NN:** PyTorch, scikit-learn, MLP, LSTM, CNN, Optuna, Feature Engineering, InstructIR, Behaviour Modelling
- **Aerospace:** Neural IOD, CR3BP Dynamics, Trajectory Simulation, Angle-Only Processing, Gauss/Laplace Methods
- **Languages & Tools:** Python, C#, JS, SQL, HTML/CSS, Django, Git, VS Code, Jupyter, Google Colab
- **Expanding:** PINNs, Unreal Engine 5, Multiplayer Architecture

Languages

- English: Fluent (C1)
- Deutsch: B1 → B2
- Hindi & Gujarati: Native

Hobbies & Soft Skills

- Archery · Swimming · Sports
- Goal-oriented · Analytical · Teamwork